Research Methodology

Week 5 Assignment: Questions, Answers and Explanations

Week 5: Creativity in Research and Ethics in Research

NPTEL course noc26_ge79 (SWAYAM), 10 questions, 1 point each

Correct answers are the NPTEL portal's accepted answers. For every option the table gives a verdict and the reason, and each question ends with a short explanation of the correct answer. This week covers intelligence and creativity, stress, what makes a good researcher, and authorship ethics.

Answer key at a glance

Question	Type	Correct option(s)	Your portal score
1	Select all that apply	**A, B**	0.5 (partial: only one of the two correct options chosen)
2	Single answer	**B**	1 (full marks)
3	Single answer	**D**	1 (full marks)
4	Single answer	**A**	1 (full marks)
5	Single answer	**B**	1 (full marks)
6	Select all that apply	**A, B**	0.5 (partial: only one of the two correct options chosen)
7	Select all that apply	**B, D**	0.5 (partial: only one of the two correct options chosen)
8	Single answer	**A**	1 (full marks)
9	Select all that apply	**A, B, C, D**	1 (full marks)
10	Select all that apply	**A, B, C**	1 (full marks)

Questions with explanations

Question 1 of 10

According to Sternberg's theories of intelligence practical intelligence relates to-

Select all that apply. *Your portal score: 0.5 (partial: only one of the two correct options chosen).*

Option	Statement	Verdict	Reason
A	**How one relates to the environment around them**	Correct	Practical intelligence is adapting to and handling one's real surroundings.
B	**How street smart they are**	Correct	Street smartness is the everyday term for practical intelligence.
C	How well they come up with new theories	Wrong	Producing new ideas and theories is creative intelligence.
D	How well they solve new problems	Wrong	Handling novel problems is also creative intelligence.

Correct answer: A and B.

Explanation: Sternberg's triarchic theory has three parts: analytical intelligence (solving structured problems), creative intelligence (new ideas, new theories, new problems) and practical intelligence (adapting to the environment, being street smart). Only the first two options describe the practical part.

Question 2 of 10

Few research scholars collaborated to work on some specific area. They generated some data and analysed the results. Some of them published the result independently since they were the first to type it up. Is it ethical to do so?

Single answer.

Option	Statement	Verdict	Reason
A	Yes, because they have collaborated and know each other	Wrong	Knowing each other does not give anyone the right to publish alone.
B	**No, all authors should know and agree with the publication**	Correct	Everyone who contributed must be informed, agree and be credited.
C	Yes, if the collaborators are junior to you	Wrong	Seniority does not change the rule.
D	No, only if the authors are senior to you	Wrong	The rule applies to everyone, not only when co-authors are senior.

Correct answer: B.

Explanation: Publishing joint work without the knowledge and agreement of all contributors is unethical, whoever they are. Authorship ethics requires that all authors know about and approve the publication.

Question 3 of 10

Creativity is seen in

Single answer.

Option	Statement	Verdict	Reason
A	Drama	**Wrong**	True, but it is only one of the fields where creativity appears.
B	Science fiction	**Wrong**	True, but it is only one of the fields.
C	Engineering	**Wrong**	True, but it is only one of the fields.
D	**All of the above**	**Correct**	Creativity means producing something new and useful, and that happens in all three.

Correct answer: D.

Explanation: Creativity is not limited to the arts. It appears in drama, in science fiction and in engineering, so the complete answer is all of the above.

Question 4 of 10

A good researcher is often well read in many areas

Single answer.

Option	Statement	Verdict	Reason
A	**True**	**Correct**	Reading across areas lets a researcher connect ideas from different fields.
B	False	**Wrong**	Narrow reading limits the ideas available.

Correct answer: A.

Explanation: New research ideas often come from joining ideas of different areas, which is easier for someone who reads widely.

Question 5 of 10

We can work to the best of our potential when we are —

Single answer.

Option	Statement	Verdict	Reason
A	Hypo stressed	**Wrong**	Too little stress gives too little drive.
B	**Eustressed**	**Correct**	Eustress is positive, motivating stress that sharpens focus.
C	Hyper stressed	**Wrong**	Too much stress reduces performance.
D	Distressed	**Wrong**	Distress is harmful stress that lowers performance.

Correct answer: B.

Explanation: Performance is best under a healthy amount of challenge, called eustress. Both too little (hypo) and too much (hyper) stress, and harmful stress (distress), reduce performance.

Question 6 of 10

The measure of a successful researcher is that the work of the researcher should be

Select all that apply. *Your portal score: 0.5 (partial: only one of the two correct options chosen).*

Option	Statement	Verdict	Reason
A	**Recognized by peers at an international level**	Correct	Success is judged by the wider research community.
B	**Typically published in peer reviewed Journals**	Correct	Peer review is how the community checks and accepts research.
C	Mentioned in the campus magazine of the college	Wrong	A campus mention is not a measure of research standing.
D	Must be discussed in the Department meeting of the researcher's department	Wrong	Internal discussion is not a measure of success.

Correct answer: A and B.

Explanation: A successful researcher's work is accepted through peer-reviewed publication and recognised by peers internationally. Local mentions do not measure research success.

Question 7 of 10

Creative people are

Select all that apply. *Your portal score: 0.5 (partial: only one of the two correct options chosen).*

Option	Statement	Verdict	Reason
A	More likely to follow societal restrictions	Wrong	Creative people question restrictions, so this is the opposite.
B	**Less likely to follow societal restrictions**	Correct	They are willing to depart from conventions.
C	More likely to think within the limits of previous work in the area	Wrong	They tend to go beyond previous work.
D	**More likely to come up with a surprisingly new solution to a problem**	Correct	Questioning limits leads to unexpected solutions.

Correct answer: B and D.

Explanation: Creative people question conventions and limits, which makes them less bound by rules and more likely to reach surprising solutions.

Question 8 of 10

A PhD usually makes you

Single answer.

Option	Statement	Verdict	Reason
A	**A specialist in one narrow area**	**Correct**	A PhD goes deep into one problem and builds expert knowledge of it.
B	A person who conducts experiments fast	**Wrong**	Speed is a skill, not what the degree is designed to build.
C	A person who has a wide knowledge in a wide range of topics	**Wrong**	A PhD gives depth in one area, not breadth across many.
D	A person who makes excellent sketches	**Wrong**	Sketching is unrelated to the aim of the degree.

Correct answer: A.

Explanation: A PhD is deep, not broad. You become an expert in one narrow area by solving a problem in it.

Question 9 of 10

To become a good researcher, one must usually

Select all that apply.

Option	Statement	Verdict	Reason
A	**Have good communication skills**	**Correct**	Results must be shared through papers and talks.
B	**Have mastery over specific techniques**	**Correct**	Technique is needed to do the work.
C	**Have a line of investigation**	**Correct**	A line of investigation gives the work direction.
D	**Be aware of recent advances in the field**	**Correct**	It avoids repeating others' work and shows the gaps.

Correct answer: A and B and C and D.

Explanation: All four are needed: communication, technique, a line of investigation and awareness of the latest developments.

Question 10 of 10

The followings can be attributed to a fully professional researcher-

Select all that apply.

Option	Statement	Verdict	Reason
A	**They have some fundamental concepts that their peers (international) might be interested in**	**Correct**	Their ideas are worth the attention of international peers.
B	**They have command over their area**	**Correct**	A professional masters the field.
C	**They are aware of the situation where they can make significant contribution**	**Correct**	They know where a contribution can be made.
D	They work without focusing on literature in the area	**Wrong**	Ignoring the literature leads to repeated work and is unprofessional.

Correct answer: A and B and C.

Explanation: A fully professional researcher has ideas of international interest, command of the area and awareness of where to contribute, and always keeps track of the literature.